

Defense Transcript Review - September 26, 2026

Defense Q&A, September 2026: independent question-by-question review

Second pass, done from the raw transcript without reusing the first file. For each question: what was asked, what you said (condensed from the raw ASR), what went wrong, a better answer, and what to p

Ground rule for reading this: an examiner's question is an invitation to show you understand your own method. Every answer below that failed did so in one of four ways: (a) answered a different questi

Prof. Molenberghs

Q1 [12:42] "What does the fuzzy in fuzzy c-means refer to?"

He prefaced it with "you're the expert, forgive a naive question." That is a courtesy; it calls for a careful, respectful answer.

****You said [12:59-13:47]:**** "C-means is just a fancy way to say k-means... fuzzy refers to soft cluster centroids... each patient gets a proportion of how it belongs to each centroid. A soft label."

****Problem:**** The substance (soft membership) is right, but "a fancy way to say k-means" is flippant and inaccurate. Fuzzy c-means (Bezdek, 1981) is a different algorithm with its own objective and a f

****Better:****

****Prepare:**** one sentence per method in your comparison: what it optimizes, its main parameter, and the value you used.

Q2 [13:53, restated 14:32] "Are the practical conclusions robust across clusteri

****You said [14:56-15:40]:**** k-means is the benchmark because it's the most robust; it depends on data size; the literature uses 8 to 10k patients; with many records simpler methods are more robust.

****Problem:**** You never answered. He asked about agreement between your methods, and you have the number in your own thesis (the pairwise ARI figure). The answer about sample size is unrelated to his q

****Better:****

****Prepare:**** a backup slide with the ARI matrix and a per-cluster overlap table (for each k-means cluster, the share of its patients that land together under each other method).

Q3 [15:42] "Why is k-means the benchmark?"

****You said [15:48-16:35]:**** "That's a bad one. Personal reason is that this is the first clustering method I learned... it's like why the normal distribution is the most common... everybody uses it as

****Problem:**** "That's a bad one" sounds like you are rating his question. "The first method I learned" is not a scientific justification. The normal-distribution analogy is not an argument.

****Better:****

Q4 [16:46] "In Figure 5.4 you show pairwise ARI. What is ARI?"

****You said [17:03-17:17]:**** "It's probably like an agreement rate... or something like that... if it's closer to one it's a perfect alignment, but it never happens."

****Problem:**** This is a figure in your own thesis and you answered with "probably". This alone can decide a grade. "It never happens" is also wrong; two identical partitions give $ARI = 1$.

****Better:****

****Prepare:**** be able to define every metric that appears in your thesis (silhouette, ARI, permutation p-value, your stability statistic) in two sentences each, without notes.

Examiner 2

Q5 [17:24] "You used k-means to select $k = 5$, then compared methods at $k = 5$, an

This is the "hyperparameter optimization" question in the written feedback.

****You said [18:07-19:15]:**** Yes; ran a grid search $k = 3$ to 24 for all methods; most methods peak at $k = 2$ or 3, "not good enough"; comparing methods at different k loses a common criterion; "delibera

****Problem, in three layers:****

1. You opened with "yes, I tested it" and then described a grid search that showed the opposite: other methods preferred a different k , and you overrode that. So the honest answer was "no, and the com

Note: the written feedback says you deflected with "how k-means creates spherical clusters". In the transcript that explanation comes at 34:07, in answer to Pedro. Either the ASR dropped it here, or t

****Better (say this before being asked, in the presentation):****

****Prepare:**** actually run it: (a) k chosen by a method-agnostic stability criterion, (b) each method at its own optimum, (c) sensitivity of the five clusters to $k = 4$ and 6. Put it in a chapter, not a

Q6 [19:18] "For interpreting the clusters you used the ATC codes. Have you consi

****You said [19:39-20:20]:**** "CART doesn't work at my scale... maybe on 10k or 100k patients, not one million... I tried it... along with DBSCAN and other stuff." After she explained supervised vs unsu

****Problem:**** You answered before understanding the question, dismissed it ("doesn't work"), and confused a supervised tree with a density clustering method. For a statistics examiner, the supervised/u

****Better:****

If you don't understand a question, say "Do you mean using the cluster labels as the outcome of a supervised model?" before answering.

****Prepare:**** fit the tree, report accuracy and the top splits, and use its rules to justify the cluster names.

Q7 [21:41, followed up 22:41] "Do you have longitudinal data? Have you used long

****You said [21:56-23:05]:**** "Yes... it doesn't work... limitation of the time... literature has at least five years... you only have four seasons." On follow-up: "Limited of time. If I have five years

****Problem:**** "Limitation of time" is ambiguous; she may have heard "I ran out of time" when you meant "the observation window is one year." She asked a concrete follow-up (too much data? no trends?) a

****Better:****

****Prepare:**** one backup slide with what you tried and the ARI/stability numbers. If you didn't try it properly, say so in one sentence rather than "it doesn't work."

Q8 [23:06] "All your techniques are distance-based on continuous data. What would you do for categorical data?"

****You said [23:25-24:44]:**** the original data has categorical fields; "you're not at my midterm defense committee, back then I did one-hot encoding"; "k-means also worked for categoricals"; CLARA "is

****Problem:**** Two incorrect statements to a statistics professor: k-means is not appropriate for categorical variables, and CLARA is a subsampling scheme for k-medoids, not a categorical method. Random

****Better:****

Alain (APB)

Q9 [25:10] "If you didn't experiment with the features, what led you to these 14 features?"

****You said [25:28-26:26]:**** advice from advisors; we agree the thesis lacks an exhaustive feature construction with PCA; would do that with more time; would explore mixed features.

****Problem:**** "Advice from advisors" hands the design to someone else; you should own it. The PCA idea is weak: PCA on hand-crafted features doesn't fix feature design, and the components are hard to interpret

****Better:****

****Prepare:**** the leave-one-feature-out ARI table. It also answers his next point about distinct pharmacies.

Q10 [26:27] "I'd expect distinct pharmacies visited contributes nothing. And I'm not sure about gender."

****You said [26:38-27:59]:**** "I'm sorry. Gender?"; "I find it difficult to interpret gender, sorry, wrong word"; binary variables are hard to measure the contribution of; agree distinct pharmacies isn't

****Problem:**** A visible fumble on a simple point. The reason for excluding sex is fine (binary variables distort Euclidean distance) but was buried under apologies. You conceded distinct pharmacies contribute

****Better:****

Q11 [28:02-30:57] "Your violin plots have enormous spikes; maybe your cutoff should be higher."

****You said [28:50-30:56]:**** good point; go to the 99.99th percentile; the extreme records are "absolutely not real patients"; "unfortunately I didn't think about it"; did not identify these patients;

****Problem:**** You asserted "not real patients" with no evidence, to the person who owns the data and who was raising a patient-safety concern. Moving a percentile cutoff is arbitrary and doesn't answer the question

****Better:****

****Prepare:**** do the DDD-based check, report how many patients it affects and what they look like, and rerun with a plausibility cap instead of a percentile cap.

Pedro

Q12 [31:46] "Your variables have different scales. Did you take care of it, and

****You said [32:06-33:00]:**** standardized; log-transformed the heavily skewed ones; nothing fancier.

****Problem:**** Adequate but thin, and you didn't answer the second half (how it affects the different methods).

****Better:****

Q13 [33:05] "What would happen without standardization?"

****You said [33:10-33:40]:**** "It goes way off the table, it explodes... some features contribute a lot."

****Better:**** "The partition becomes a binning of the single highest-variance feature, total annual quantity, so the clusters are just low/medium/high volume." Say the concrete outcome, not "it explodes"

Q14 [33:46] "Are there limitations of k-means regarding the distributions you ob

****You said [34:07-35:45]:**** k-means separates in Euclidean space; spherical k-means uses cosine distance; skewed distributions need pre-processing; winsorize at 99.9/0.1; log-transform.

****Problem:**** A tour of pre-processing, not an answer about k-means's assumptions. This is possibly the "spherical clusters" answer the jury remembered as a deflection.

****Better:****

Chair

Q15 [35:59] "How did you arrive at the pharmacological descriptions, e.g. 'cardi

****You said [36:39-38:50]:**** statistics student; had a code-to-name mapping; went through the A10 category; "a lot of dirty work"; heat map shows the separation; "mapping the code to its original term"

****Problem:**** He asked a direct yes/no question about AI assistance and you didn't answer it. He named the terms because they are not a statistics student's vocabulary, so he already suspected the answ

****Better:****

KU Leuven's [guidelines on authorised GenAI use](<https://www.kuleuven.be/english/education/student/educational-tools/guidelines-for-students-on-the-authorised-use-of-genai>) expect disclosure. Whatever

Laughter and disrespect: second, independent pass

Two signals, checked separately.

Signal 1: ASR hallucination markers on your channel

Speech recognizers produce filler phrases or lone punctuation on your channel when they pick up non-speech sound: laughter, a sigh, an exhale. Your channel shows this at these points, all while an exa

| Time | ASR output on your line | What the examiner was saying |

Treat this as a hypothesis. But 28:38-29:31 is the strongest candidate for laughter in the whole recording, and it is the worst place for it: the pharmacists' representative was raising a patient-safe

Signal 2: wording

Ranked by how likely each is to be what the jury meant by "disrespectful."

1. ****19:02-19:15****, "it's more like a rhetorical one... this is flawless... I admit, a flaw." Sarcastic concession on the key methodological question.

The pattern behind all of it

Every flagged moment is the same reflex: when a question exposes a weakness, you defuse the tension with a joke, a personal aside, or a concession delivered lightly. In a defense that reads as not tak

1. Two-second pause. No laugh, no "thank you for asking."

If you don't know: "I don't know. I'll check and add it." Nothing after that.

Before January, in priority order

1. Fair method comparison (Q5): method-agnostic choice of k , each method at its own optimum, cross-method ARI, sensitivity at $k = 4$ and 6 .